Skanda Bharadwaj

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Education

Pennsylvania State University
Ph.D., M.Sc. in Computer Science and Engineering — Aug 2021 – Present; Jan 2019 - May 2021

PES Institute of Technology
B.Tech in Telecommunications Engineering — Aug 2011 – May 2015

Skills & Specialization

Computer Vision, Deep Learning, Recommendation Systems, Content Understanding, PyTorch

Work Experience

Machine Learning Engineer Intern @ Meta – Menlo Park, CA — May - Aug 2026

- Optimized creator experience for Facebook Reels by improving cold-start content distribution through large-scale retrieval, ranking, and ML-driven VPV budget allocation, contributing to an 11pp statistically significant increase in creator sessions.
- Designed and implemented an end-to-end LightGBM-based ML system processing 170M creators, spanning feature engineering, training, serving infrastructure, and online experimentation to inform personalized VPV budget allocation.

Machine Learning Engineer Intern @ Meta – Bellevue, WA — May - Aug 2025

- Designed and implemented recommendation pipelines to improve cold-start content discovery for emerging Facebook Reels creators using KNN, LLM embeddings, and creator clustering.
- Expanded recommendation coverage to >99.9% for 26K+ newly onboarded creators in offline evaluation, leading to integration into Meta's production recommendation system.

Research Scientist Intern @ SLB-Doll Research – Cambridge, MA — May – Aug 2024

- Developed a real-time image processing pipeline that integrates traditional techniques with deep learning classifiers to segment, detect, and localize methane plumes for enhanced environmental monitoring.

Computer Vision Engineer @ Continental – Bengaluru, India — July 2015 – Dec 2018

- Played a key role in the development of multi-object tracking For Advanced Driver Assistance Systems (ADAS).

Research

Research Assistant @ LPAC — Aug 2021 – Present

- Developing machine learning and computer vision methods that leverage human perceptual grouping and recurring visual patterns to learn robust visual representations for scene understanding.
- Developing interpretable multi-modal vision models for geometric scene understanding and human stability estimation using image geometry, human pose, and foot-pressure dynamics.

Research Assistant @ UIT Lab — Jan 2019 – May 2021

- Developed Siamese CNN and Kalman filter–based methods for robust ultrasound image tracking, improving motion estimation in medical imaging.

Selected Publications **Google Scholar**

- **Skanda Bharadwaj**, R. T. Collins and Y. Liu, "Recurrence-based Vanishing Point Detection", In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* February 2025, pp. 8909–8918. [Paper]
- K. Y. Kraiger, J. Li, **Skanda Bharadwaj**, Y. Liu, and R. T. Collins, "Estimating Foot Pressure and Stability from Visual Input", In *36th British Machine Vision Conference 2025, BMVC*, Sheffield, UK, November 24-27, 2025. [Paper]
- Shimian Zhang, **Skanda Bharadwaj**, Keaton Kraiger, Yashasvi Asthana, Hong Zhang, Robert Collins, and Yanxi Liu, "Novel 3D Scene Understanding Applications From Recurrence in a Single Image", *arXiv preprint* 2022. [Paper]
- **Skanda Bharadwaj**, Sumukha Prasad, Mohamed Almekkawy, "An Upgraded Siamese Neural Network for Motion Tracking in Ultrasound Image Sequences", *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 68(12):3515-3527, 2021. [Paper]